kiosk-admin

RDP Browser-Kiosk Server with Web Management Console
Installation & User Guide

Version 1.8.5 · July 2026

1. What is kiosk-admin?

kiosk-admin turns a single Ubuntu/Debian machine into a multi-user **RDP kiosk server**. Users connect with any standard RDP client (Windows *Remote Desktop*, Remmina, FreeRDP, mobile RDP apps) and receive a **locked-down browser with a fixed set of tabs — 5 by default**. The tab strip at the top of the screen is the **only** visible browser UI: no address bar, no menus, no settings, no developer tools, no downloads. The default tabs are permanent (no close button); links that open in a new tab appear *after* them with a normal close button, so users can dismiss those when done. If the browser crashes or is closed, it restarts by itself within seconds.

Administrators manage everything from a web console:

- **Live dashboard** — every logged-in user shown online, one click from a live view of their screen
- **Screen mirroring & remote control** — click any session to watch it live or take over mouse and keyboard (noVNC), with a “view only” toggle
- **Hard refresh** — send Ctrl+Shift+R to a user's browser remotely
- **Browser reset** — kill and relaunch a user's browser with a clean profile (clears cookies, history, and any wedged state)
- **User management** — create kiosk users, set passwords, delete users
- **Session logout** — terminate any user's RDP session

Architecture at a glance

```
RDP clients ──> xrdp ──> per-user Xorg session (:10, :11, ...)
                              ├──> Firefox, tab-strip-only UI  (respawn loop = self-healing)
                              └──> x11vnc on 127.0.0.1:59<display>  (mirroring)

Admin browser ──> kiosk-admin web console (HTTPS, port 8443)
                     ├── session list, screenshots, user management
                     └── websocket bridge ──> x11vnc ──> live view & control
```

Security model: the web app runs as an unprivileged user; every privileged action goes through a single audited root helper (`kiosk-ctl`) allowed by one sudoers line. VNC traffic never leaves localhost — the only network entry points are RDP (3389) and the password-protected admin console (8443, TLS).

2. Requirements

Item	Requirement
Operating system	Ubuntu 22.04 / 24.04 or Debian 12 (server or desktop)
Access	Root (sudo) on the target machine
Network	Internet access during install (apt packages); ports 3389 and 8443 reachable by clients
Hardware	Roughly 400–600 MB RAM per concurrent kiosk user (Firefox with 5 tabs); 2 GB + 600 MB/user is a safe sizing rule

The installer pulls everything else automatically: xrdp, xorgxrdp, openbox, x11vnc, xdotool, ImageMagick, noVNC, Firefox, and a private Python environment.

3. Installation

1. Copy this repository to the target machine (git clone, scp, or unpack the release tarball):

   ```
   git clone <repo-url> kiosk-admin
   cd kiosk-admin
   ```

2. Run the installer:

   ```
   sudo bash deploy/install.sh
   ```

   (`make install` is an equivalent shortcut if `make` is installed; fresh servers usually don't have it)

3. At the end, the installer prints the console address and a **generated admin password**:

   ```
   == done: kiosk-admin 1.0.0 ==
      admin console : https://192.168.1.50:8443
      admin password: Xy9kQ2mP4LcA   (change with: sudo kiosk-admin-passwd)
   ```

> **Save the admin password now.** It is shown only once. You can replace it at any time with `sudo kiosk-admin-passwd`.

3.1 First-time setup

1. **Set the kiosk tabs.** Edit `/etc/kiosk/kiosk.conf` — one URL per tab, space-separated, opened left to right (5 by default):

   ```
   KIOSK_URLS="https://erp.example.com https://mail.example.com https://wiki.example.com https://crm.example.com
   https://reports.example.com"
   ```

   The change applies the next time each user's browser starts — use the console's **Reset** button to apply it to running sessions immediately.

2. **Open the firewall** (if enabled) for the two service ports only:

   ```
   sudo ufw allow 3389/tcp    # RDP for users
   sudo ufw allow 8443/tcp    # admin console
   ```

3. **Log in to the console** at `https://<server-ip>:8443`. The TLS certificate is self-signed, so your browser shows a one-time warning — accept it (or install real certificates, see §7).
4. **Create the first user** with the **+ Add user** button (username: lowercase letters, digits, `-`, `_`; password: minimum 6 characters).
5. **Test the user login:** from any PC, open an RDP client, connect to the server's address, and sign in with the new username and password. The browser should fill the screen with the configured tabs across the top and no other UI — and a new session card should show up on the admin dashboard within a few seconds.

4. Using the admin console

4.1 Dashboard

The main page shows one card per active RDP session with the username, an **online** badge, display number, session uptime, and a green/red dot for browser health. Cards deliberately show no live preview — that keeps the dashboard nearly free of bandwidth and server load even with many sessions; click a card whenever you want to see the actual screen.

Button	What it does	When to use it
Refresh	Sends Ctrl+Shift+R to the user's browser (cache-bypassing page reload)	The page shows stale data or a deployment just went out
Reset	Kills the browser; it relaunches within ~2 seconds with a brand-new profile (cookies, history, and session state wiped; saved website passwords are kept) and the full default tab set restored	The browser is frozen, logged into the wrong account, tabs were lost, or the tab list in `kiosk.conf` changed
Logout	Terminates the user's whole RDP session	Freeing a seat, or ending an abandoned session

4.2 Screen mirroring & remote control

1. Click a session card (or open it in a new tab). The live viewer connects in about a second.

2. By default **you control the session**: your mouse and keyboard act on the user's screen — ideal for guided support.
3. Tick **view only** in the toolbar to watch without interfering.
4. Tick **touch screen** if *you* are mirroring from a touch device: toolbar controls grow to finger size and a dot marks the remote cursor so taps have visible feedback. It defaults to on when a touch device is detected and is remembered per browser.
5. Admins also see **user's touch screen** — the kiosk-side setting for the user being viewed (hidden mouse pointer, finger-sized tabs). Toggle it, then press **Reset browser** to apply immediately.
6. The viewer toolbar also has **Refresh page** and **Reset browser** buttons for the session being viewed.

> The user keeps full control of their own session at all times — mirroring is shared, not exclusive. Both sides see the same screen.

4.3 Managing users

- **Add:** "+ Add user" in the header. The account works for RDP immediately. Tick **Touch screen** for users on touch terminals: their session hides the mouse pointer and gets finger-sized tabs.
- **Touch screen toggle:** the checkbox in the *Kiosk users* table (also available in the live viewer's toolbar) switches an existing user between touch and mouse mode — it applies on **Reset** or at their next login.
- **Set password:** per-user button in the *Kiosk users* table.
- **Delete:** per-user button; logs the user out first, then removes the account **and its home directory**. This cannot be undone.
- The table also shows who is currently online.

> Only accounts in the `kioskusers` group are visible to (and manageable by) the console. Regular admin accounts on the same machine are untouched and still get a normal desktop over RDP.

4.4 Console users & roles

The *Console users* table manages accounts for the web interface itself (not RDP users). Two roles exist:

Role	Can	Cannot
admin	Everything	—
viewer	See the dashboard and open the live view to watch or control any user's screen	Add/delete kiosk or console users, change passwords, refresh/reset browsers, log sessions out

Restrictions are enforced by the server, not just hidden in the UI. The built-in `admin` account cannot be deleted; its password is managed on the server with `sudo kiosk-admin-passwd`. Additional accounts are stored in `/var/lib/kiosk-admin/webusers.json` and survive upgrades.

> Give support staff **viewer** accounts: they can help users hands-on via screen control, but cannot change the system.

5. Upgrading

Upgrading **is** reinstalling — the installer is idempotent:

```
cd kiosk-admin
git pull                     # or unpack the new release tarball over this directory
sudo bash deploy/install.sh  # or: make upgrade
```

Survives every upgrade	Replaced by the upgrade
`/etc/kiosk/kiosk.conf` (kiosk URL) Admin password, cookie secret, TLS certs All kiosk user accounts and home dirs **Live RDP sessions** (xrdp is never restarted)	Web app in `/opt/kiosk-admin/` Scripts in `/usr/local/bin/` `/etc/xrdp/startwm.sh`, Chromium policies systemd unit, Python dependencies

Only the admin console restarts (1–2 seconds); users notice nothing. Updated session scripts take effect at each user's next login. To upgrade an offline server: on any machine run `make package`, copy the resulting `kiosk-admin-<version>.tar.gz` across, unpack, `make install`.

6. Uninstalling

```
sudo bash deploy/uninstall.sh           # removes app + scripts, KEEPS configs
sudo bash deploy/uninstall.sh --purge   # also deletes /etc/kiosk and /etc/kiosk-admin
```

Kiosk user accounts are never deleted automatically — remove them from the web UI beforehand, or afterwards with `sudo userdel -r <name>`. xrdp itself stays installed; the original `startwm.sh` is restored.

7. Configuration reference

/etc/kiosk/kiosk.conf

Setting	Default	Meaning
`KIOSK_URLS`	5 example sites	The user's tabs: one URL per tab, space-separated, opened left to right. The tab strip is the only browser UI users see.
`KIOSK_FRESH_PROFILE`	no	`no`: keep cookies, so users **stay signed in to websites** across browser restarts and RDP logins/logouts. `yes`: wipe the browser profile on every (re)start — users must sign in to websites again after every RDP login. Admin **Reset** forces a full wipe either way.
`KIOSK_SAVE_LOGINS`	yes	`yes`: Firefox offers to save website passwords (the “Save login?” popup appears just below the tab strip after signing in), and saved logins survive profile wipes, browser restarts, and admin Resets. `no`: password saving is disabled and existing saved logins are wiped on the next restart.
`KIOSK_RESTART_DELAY`	2	Seconds before relaunching a closed/crashed browser
`KIOSK_BG`	#1a1a2e	Desktop color visible briefly before the browser starts
`KIOSK_NUMLOCK`	on	NumLock state inside the session at login: `on`, `off`, or `leave`. RDP starts the session out of sync with the client keyboard, making NumLock appear inverted; pinning it avoids that.
`KIOSK_TOUCH_MODE`	no	`yes`: touch-screen mode for **all** users — the mouse pointer is hidden on the kiosk screen, Firefox gets touch-friendly input handling and pinch zoom, and the tab strip grows to finger size. Individual users can be switched to touch mode regardless of this setting via the **Touch screen** checkbox in the console (membership in the `kiosktouch` group). Admin mirroring still works normally.

Other locations

Path	Purpose
`/etc/kiosk-admin/admin.passwd`	Built-in admin password (bcrypt); change via `sudo kiosk-admin-passwd`
`/var/lib/kiosk-admin/webusers.json`	Additional console accounts (admin/viewer roles), managed from the web UI
`/etc/kiosk-admin/certs/`	TLS cert/key. Replace `cert.pem` / `key.pem` with real ones, then `sudo systemctl restart kiosk-admin`
`/opt/kiosk-admin/`	Installed web app (replaced on upgrade — never edit here)
`~<user>/.kiosk/`	Per-user logs: browser.log, x11vnc.log, session.log

Ports

Port	Service	Exposure
3389/tcp	xrdp — user logins	Open to users
8443/tcp	Admin console (HTTPS + WebSocket)	Open to admins only, ideally restricted by firewall source IP
59xx/tcp	x11vnc per session	127.0.0.1 only — never exposed

8. Troubleshooting

Symptom	Cause & fix
Console unreachable on :8443	Check `systemctl status kiosk-admin` and `journalctl -u kiosk-admin -e`; confirm the firewall allows 8443.
User gets a black/empty RDP screen	Browser may be starting (snap Firefox's first launch per user can take up to a minute); check `~<user>/.kiosk/browser.log`. Verify Firefox is installed and the `KIOSK_URLS` entries are valid.
browser.log: “Authorization required ... cannot open display”	Snap Firefox cannot read `~/.Xauthority`. Fixed in v1.3.1 (the session grants X access via `xhost +si:localuser`) — upgrade, then log the user out and back in.
Fonts look wrong / jagged, or text shows as boxes	Server installs ship almost no fonts. The installer pulls a proper set since v1.3.2 (Liberation, DejaVu, Noto core + emoji, Indic); for other languages add e.g. `fonts-noto-cjk`. After installing fonts, press **Reset** on each session.
My site needs a specific font (e.g. a Google Font)	Drop the `.ttf` / `.otf` into `system/fonts/` in the repo and re-run the installer — it lands in `/usr/local/share/fonts/kiosk-admin/`. Fredoka ships bundled since v1.6.1. Reset sessions to apply.
“no active session” on Refresh/Screenshot	The user has no Xorg process — they disconnected and the session was reaped. They simply need to reconnect.
Live viewer says “connection lost”	Upgrade to v1.3.3+ first (older versions had a websocket handshake bug that broke the viewer with modern noVNC). Otherwise: console login expired (sign in again on the dashboard) or the session ended. Server log: `journalctl -u kiosk-admin`; x11vnc log: `~<user>/.kiosk/x11vnc.log`.
Duplicate sessions for one user	Reconnects with a different color depth create a second session. Set `Policy=Default` and `KillDisconnected=false` in `/etc/xrdp/sesman.ini`, restart xrdp off-hours.
The “Save login?” popup never appears when signing in to a website	Upgrade to v1.8.1+: older versions hid the toolbar the popup anchors to, which silently suppressed it. After upgrading, press **Reset** on the session, sign in to the site again, and the popup appears just below the tab strip. If a user once tapped “Never save”, a **Reset** also clears that choice (already-saved logins are kept).
An address bar / “Not secure” banner pops up over the page	Upgrade to v1.8.5+ and press **Reset**. In v1.8.1–v1.8.4 a tap at the very top screen edge (or Ctrl+L) could focus the invisible address bar, which then expands over the page. The “Not secure” chip itself means that tab's URL is `http://` or has a certificate problem — use `https://` URLs in `KIOSK_URLS` where possible.
Touch user still sees the mouse pointer when touching the screen	Upgrade to v1.8.4+ and press **Reset** on the session. Older versions hid the pointer only while it was idle — and every touch moves the pointer, un-hiding it at exactly the wrong moment. Touch mode now switches the session to a fully transparent cursor theme, so the pointer is invisible even while moving (including over RDP).
Websites ask for their password again after every RDP login	Website sessions live in cookies, and `KIOSK_FRESH_PROFILE=yes` wipes them on every browser start. Since v1.8.2 the default is `no` (users stay signed in) — but upgrades never touch an existing `/etc/kiosk/kiosk.conf`, so on older installs set `KIOSK_FRESH_PROFILE="no"` there yourself. The **Reset** button still wipes everything when needed.
NumLock appears	The RDP client and the session start out of sync. Since v1.3.4 the session pins NumLock on at login

inverted (LED on = keypad acts off)	(`KIOSK_NUMLOCK`). If it drifts mid-session, pressing NumLock twice resyncs it.
Wrong admin password / lost password	`sudo kiosk-admin-passwd` — effective immediately.
Tab list change not visible	`KIOSK_URLS` is read at browser start. Press **Reset** on each session (or wait for the users' next login).
A user closed some of their default tabs	Default tabs have no close button, but Ctrl+W still works. Press **Reset** to restore the full default tab set. (Tabs opened from links are meant to be closeable — that is by design.)